§ Principle of Inclusion-Exclusion (PIE) Let $\mathcal{U}$ be a universal set for a counting problem. Let $C_1, C_2, \ldots, C_n$ be a finite collection of finite sets where each $C_i \subseteq \mathcal{U}$. The intersection of all C_i defines a condition set for the cardinality, C, as follows.

$$C = \bigcap_{i \in [n]} C_i = C_1 \cap C_2 \cap \ldots \cap C_n$$

and the complementary sets $\overline{C_i} = \mathcal{U} \setminus C_i$ for $i \in [n] = \{1, 2, \ldots, n\}$ define the sets of elements that do not satisfy the collection of conditions, C. Then, the number of elements in $\mathcal{U}$ that satisfy all the conditions in C is given as follows.

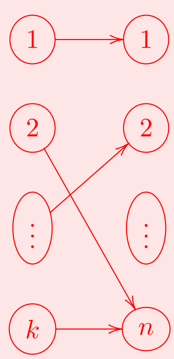
$$\begin{aligned} |C| = \left|\bigcap_{i=1}^{n} C_i\right| &= |U| - \left|\bigcup_{i=1}^{n} \overline{C_i}\right| = \sum_{I \subseteq [n]} (-1)^{|I|} \left|\bigcap_{i \in I} \overline{C_i}\right| \\ &= |U| - \sum_{i=1}^{n} |\overline{C_i}| + \sum_{1 \le i < j \le n} |\overline{C_i} \cap \overline{C_j}| - \ldots + (-1)^n |\overline{C_1} \cap \overline{C_2} \cap \ldots \cap \overline{C_n}| \end{aligned}$$

where the intersection over an empty index set, $\bigcap_{i \in \emptyset} \overline{C_i}$, is defined as the universal set U.

§ Cardinality of Surjective Functions The number of surjective functions from a set $[k] = \{1, 2, \ldots, k\}$ to a set $[n] = \{1, 2, \ldots, n\}$ is given by the formula:

$$\mathcal{S}(k, n) = \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)^k$$

Proof: By the definition of surjective functions, a function $f : [k] \to [n]$ is surjective if and only if for every $i \in [n]$, there exists at least one $j \in [k]$ such that $f(j) = i$. Let C_i be defined as follows.



C_1: there exists a value in $[k]$ that maps to 1 in $[x]$

C_2: there exists a value in $[k]$ that maps to 2 in $[x]$

C_i: there exists a value in $[k]$ that maps to i in $[x]$

Then, $|C|$ is the number of surjective functions by definition. Then, by Principle of Inclusion-Exclusion (PIE), the number of surjective functions is given as follows.

$$\begin{aligned} |C| &= n^k - \binom{n}{1}(n-1)^k + \binom{n}{2}(n-2)^k - \ldots + (-1)^n \binom{n}{n}(n-n)^k \\ &= \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)^k \end{aligned}$$

since $\sum_{i=1}^{n} |\overline{C_i}|$ is the number of functions from $[k]$ to $[n]$ such that a single value, i, is not in the range for each $i \in [n]$, which is given by $\binom{n}{1}(n-1)^k$. With similar reasoning, $\sum_{1 \le i < j \le n} |\overline{C_i} \cap \overline{C_j}| = \binom{n}{2}(n-2)^k$ and so on.

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Trig Graphs — Practice

For each function below: (i) state the period, (ii) state the phase shift, (iii) state the vertical shift, (iv) determine the domain, (v) determine the range, (vi) list vertical asymptotes in one period, and (vii) sketch one period showing asymptotes.

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1. $f(x) = -2\csc(4x) - 3$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
2. $g(x) = 3\sec\left(\frac{1}{2}x + \pi\right) + 1$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
3. $h(x) = \frac{1}{2}\tan\left(2x - \frac{\pi}{4}\right) - 2$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
4. $k(x) = -\cot(3x) + 4$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
5. $m(x) = 2\csc\left(x - \frac{\pi}{6}\right)$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
6. $n(x) = -\sec(5x + \pi) + 2$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
7. $p(x) = \tan\left(\frac{1}{3}x - \frac{\pi}{2}\right) - 1$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:
8. $q(x) = -3\cot\left(4x + \frac{\pi}{3}\right) + 1$
   - Period:
   - Phase shift:
   - Vertical shift:
   - Domain:
   - Range:
   - Asymptotes:

Trig Properties — Practice

Use trig identities to simplify the following.

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1. $\sin^2(x) + \cos^2(x)$
2. $1 + \tan^2(x)$
3. $\sec^2(x) - 1$
4. $1 - \sin^2(x)$
5. $\csc^2(x) - \cot^2(x)$
6. $\sec x - \tan x \sin x$
7. $\dfrac{\sec\theta \sin\theta}{\tan\theta + \cot\theta}$
8. $\dfrac{\sec\theta}{\cos\theta} - \dfrac{\tan\theta}{\cot\theta}$
9. $\dfrac{\sec^2\theta}{\sec^2\theta - 1}$
10. $\csc^2\theta \tan^2\theta - 1$
11. $2\cos\theta \tan\theta \csc\theta$
12. $6\cos\theta\left(\dfrac{1}{\cos\theta} - \dfrac{\cot\theta}{\csc\theta}\right)$
13. $\dfrac{\tan x + \sin x \sec x}{\csc x \tan x}$
14. $\csc x \sec x - \tan x$
15. $\dfrac{1 - \sin^2 x}{\csc^2 x - 1}$
16. $\csc x - \cos x \cot x$
17. $\sec x \cot x - \sin x$
18. $\dfrac{\csc x \cos x + \cot x}{\sec x \cot x}$
19. $\dfrac{\sec^2 x}{\cot^2 x + 1}$
20. $\dfrac{\sec x \csc x - \tan x}{\sec x \csc x}$
21. $\cot x - \csc^2 x \cot x$
22. $\cot x - \cos^3 x \csc x$
23. $\dfrac{\cos \dot{x}}{\sec x + 1} + \dfrac{\cos x}{\sec x - 1}$
24. $\dfrac{1}{\sec x + 1} + \dfrac{1}{\sec x - 1}$
25. $\dfrac{1 - \cos x}{\tan x} + \dfrac{\sin x}{1 + \cos x}$
26. $\dfrac{\cos x \cot x}{\sec x + \tan x} + \dfrac{\sin x}{\sec x - \tan x}$
27. $\dfrac{\sin x}{\csc x + 1} + \dfrac{\sin x}{\csc x - 1}$
28. $\tan x - \csc x \sec x$
29. $\cos x + \tan x \sin x$
30. $\csc x \tan^2 x - \sec^2 x \csc x$
31. $\sec x \csc x - \cos x \csc x$

Part 1

Solve each equation.

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1. $6 = \frac{a}{4} + 2$
2. $9x - 7 = -7$
3. $-4 = \frac{r}{20} - 5$
4. $-6 + \frac{x}{4} = -5$
5. $0 = 4 + \frac{n}{5}$
6. $-1 = \frac{5+x}{6}$
7. $\frac{v+9}{3} = 8$
8. $-9x + 1 = -80$
9. $-2 = 2 + \frac{v}{4}$
10. $2(n+5) = -2$
11. $-6 = \frac{n}{2} - 10$
12. $144 = -12(x+5)$
13. $8n + 7 = 31$
14. $\frac{n+5}{-16} = -1$
15. $-10 = 10(k-9)$
16. $-9x - 13 = -103$
17. $-10 = -10 + 7m$
18. $\frac{m}{9} - 1 = -2$
19. $-243 = -9(10+x)$
20. $9 + 9n = 9$
21. $7(9+k) = 84$
22. $8 + \frac{b}{-4} = 5$

Part 2

Solve each equation.

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1. $\frac{r}{10} + 4 = 5$
2. $\frac{n}{2} + 5 = 3$
3. $3p - 2 = -29$
4. $1 - r = -5$
5. $\frac{k - 10}{2} = -7$
6. $\frac{n - 5}{2} = 5$
7. $-9 + \frac{n}{4} = -7$
8. $\frac{9 + m}{3} = 2$
9. $\frac{-5 + x}{22} = -1$
10. $4n - 9 = -9$
11. $\frac{x + 9}{2} = 3$
12. $\frac{-12 + x}{11} = -3$
13. $\frac{-4 + x}{2} = 6$
14. $-5 + \frac{n}{3} = 0$
15. $\frac{p}{4} + 8 = 7$
16. $9 + \frac{n}{4} = 15$
17. $6 + \frac{x}{2} = 4$
18. $\frac{b + 11}{3} = -2$
19. $\frac{a - 10}{3} = -4$
20. $-12r + 4 = 100$
21. $\frac{m}{16} - 9 = -8$
22. $-7 + 4r = -15$
23. $\frac{m - 13}{2} = -8$
24. $-5x + 13 = -17$
25. $\frac{k + 10}{-2} = 5$
26. $\frac{p + 8}{-2} = 10$
27. $-14r - 19 = 303$
28. $\frac{x}{-4} - 5 = -8$

Part 3

Solve each equation.

1. $5x + 3 = 23$
2. $\dfrac{x-4}{5} = 2$
3. $-3(2x-5) = 9$
4. $7 - 2x = -9$
5. $\dfrac{3x+1}{4} = 5$
6. $0.5x - 2 = 3$
7. $4(x+2) = 3x + 14$
8. $-\frac{2}{3}x = 8$
9. $\dfrac{2x-3}{6} = -1$
10. $9 = \dfrac{x}{-3} + 4$
11. $2(3x-4) = 16$
12. $\dfrac{x}{5} + 7 = 10$
13. $-4 + \dfrac{x}{2} = 6$
14. $5 - 2x = 11$
15. $3(x-4) + 2x = 18$
16. $\dfrac{x+3}{-4} = 2$
17. $0 = -5 + 2n$
18. $\dfrac{6x+9}{3} = 7$
19. $-7 = 2(-3 + x)$
20. $\dfrac{4x}{7} = 12$
21. $-3(x+5) = 12$
22. $\dfrac{x}{4} - 3 = -8$
23. $2x + 1 = -3x + 11$
24. $\dfrac{7-x}{2} = 5$

Part 4

Solve each equation or system.

1. $0x = 5$
2. $0x = 0$
3. $2(x+1) = 2x + 3$
4. $3(x-2) = 3x - 6$
5. $\begin{cases} x + y = 2 \\ 2x + 2y = 5 \end{cases}$
6. $\begin{cases} x + y = 2 \\ 2x + 2y = 4 \end{cases}$
7. $\begin{cases} x + 2y = 3 \\ 2x + 4y = 6 \end{cases}$
8. $\begin{cases} x + 2y = 3 \\ 2x + 4y = 7 \end{cases}$

Part 5

Solve each system of equations.

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1. $\begin{cases} x + y = 5 \\ x - y = 1 \end{cases}$
2. $\begin{cases} 2x + 3y = 7 \\ 4x - y = 1 \end{cases}$
3. $\begin{cases} 3x - 2y = 4 \\ 6x - 4y = 8 \end{cases}$
4. $\begin{cases} 3x - 2y = 4 \\ 6x - 4y = 9 \end{cases}$
5. $\begin{cases} 5x + y = 11 \\ -x + 2y = 1 \end{cases}$
6. $\begin{cases} x + y = 2 \\ x + y = 3 \end{cases}$
7. $\begin{cases} a + 2b = 4 \\ 3a + 6b = 12 \end{cases}$

Part 6: 2 or more steps of inequalities

Solve each inequality.

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1. $2x - 3 < 5$
2. $3(x-1) > 2(x+2)$
3. $5 - 2x > 3x + 1$
4. $-3(2x-5) \leq 9$
5. $\dfrac{x-4}{5} \geq 2$
6. $7 - 2x < -9$
7. $\dfrac{3x+1}{4} \leq 5$
8. $0.5x - 2 < 3$
9. $4(x+2) \geq 3x + 14$
10. $-\frac{2}{3}x < 8$
11. $\dfrac{2x-3}{6} \leq -1$
12. $9 < \dfrac{x}{-3} + 4$
13. $2(3x-4) \geq 16$
14. $\dfrac{x}{5} + 7 \leq 10$
15. $-4 + \dfrac{x}{2} > 6$
16. $5 - 2x \leq 11$
17. $3(x-4) + 2x < 18$
18. $\dfrac{x+3}{-4} \geq 2$
19. $0 \leq -5 + 2n$
20. $\dfrac{6x+9}{3} < 7$
21. $-7 \geq 2(-3 + x)$
22. $\dfrac{4x}{7} > 12$
23. $-3(x+5) \leq 12$
24. $\dfrac{x}{4} - 3 \geq -8$
25. $2x + 1 < -3x + 11$
26. $\dfrac{7-x}{2} \geq 5$

Answer: Here are the answers.

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1. $x < 4$
2. $x > 7$
3. $x < \dfrac{4}{5}$
4. $x \leq 4$
5. $x \geq 14$
6. $x > 8$
7. $x \leq \dfrac{19}{3}$
8. $x < 10$
9. $x \geq 6$
10. $x < -12$
11. $x \leq -\dfrac{3}{2}$
12. $x < -15$
13. $x \geq 4$
14. $x \leq 15$
15. $x > 20$
16. $x \geq -3$
17. $x < 6$
18. $x \leq -11$
19. $n \geq \dfrac{5}{2}$
20. $x < 2$
21. $x \leq -\dfrac{1}{2}$
22. $x > 21$
23. $x \leq -9$
24. $x \geq -20$
25. $x \leq 12$
26. $x < 2$
27. $x \leq -3$

Part 7: More Inequalities

Solve each inequality or system.

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1. $-4 \leq 2x + 1$
2. $3 - x \leq 4$
3. $\dfrac{x+5}{3} > 2$
4. $2 - \dfrac{x}{4} \leq 5$
5. $-3 \leq \dfrac{2x-1}{3}$
6. $\dfrac{5x+2}{-5} \geq -3$
7. $\dfrac{2x+3}{5} \geq 1$
8. $-2(3 - x) < 6$

Part 8: Rate of Change from a Table

For each table, find the rate of change and starting value, then write the equation in the form $y = mx + b$.

X	Y
0	3
1	7
2	11
3	15

What is the rate? ____________
What is the starting value? ____________
Write the equation: ________________________

X	Y
0	-2
1	4
2	10
3	16

What is the rate? ____________
What is the starting value? ____________
Write the equation: ________________________

X	Y
0	-4
1	-2
2	0
3	2

What is the rate? ____________
What is the starting value? ____________
Write the equation: ________________________

X	Y
0	28
1	23
2	18
3	13

What is the rate? ____________
What is the starting value? ____________
Write the equation: ________________________

X	Y
0	30
1	28
2	26
3	24

What is the rate? ____________
What is the starting value? ____________
Write the equation: ________________________

X	Y
0	
1	
2	
3	

What is the rate? ____________
What is the starting value? ____________
Write the equation: $y = 2x + 3$

X	Y
0	0
1	6
2	12
3	18

What is the rate? ____________
What is the starting value? ____________
Write the equation: ________________________

X	Y
0	
1	
2	
3	

What is the rate? ____________
What is the starting value? ____________
Write the equation: $y = 3x$
